

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:19

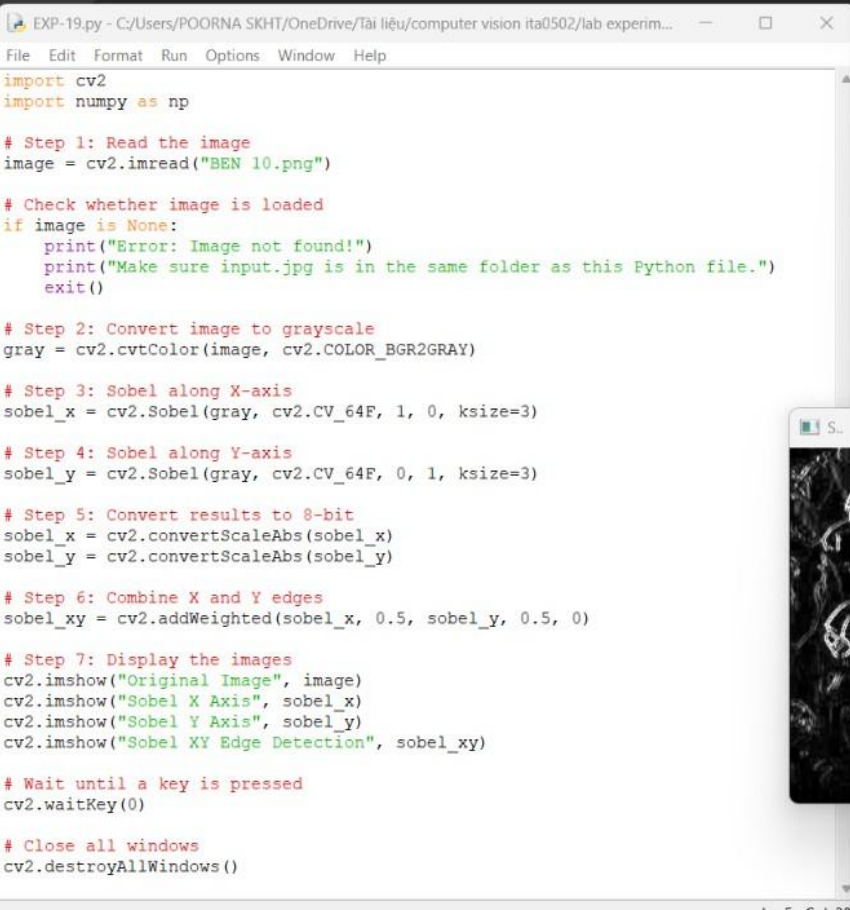
Aim:

To detect edges in an image using the Sobel operator along both the X-axis and Y-axis.

Algorithm

1. Read the input image.
2. Convert the image into grayscale.
3. Apply Sobel operator along the X-axis.
4. Apply Sobel operator along the Y-axis.
5. Combine X and Y gradients to obtain the final edge image.
6. Display the original and edge-detected images.

Code:



```
EXP-19.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
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import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure input.jpg is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Sobel along X-axis
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

# Step 4: Sobel along Y-axis
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Step 5: Convert results to 8-bit
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Step 6: Combine X and Y edges
sobel_xy = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

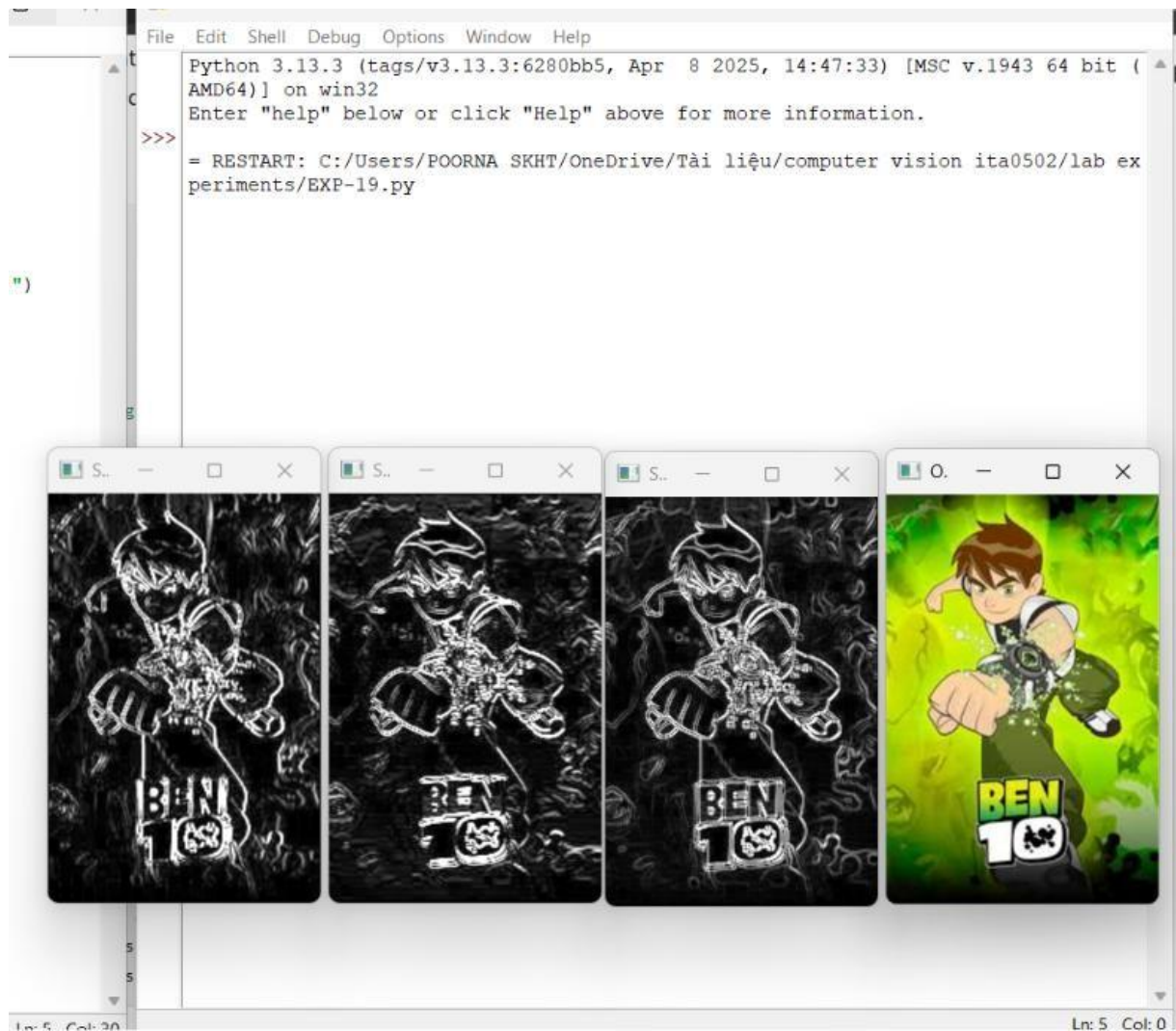
# Step 7: Display the images
cv2.imshow("Original Image", image)
cv2.imshow("Sobel X Axis", sobel_x)
cv2.imshow("Sobel Y Axis", sobel_y)
cv2.imshow("Sobel XY Edge Detection", sobel_xy)

# Wait until a key is pressed
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()
```

Ln: 5 Col: 30

OUTPUT :

**Result:**

The edges of the image are successfully detected using the Sobel operator along the X and Y axes.